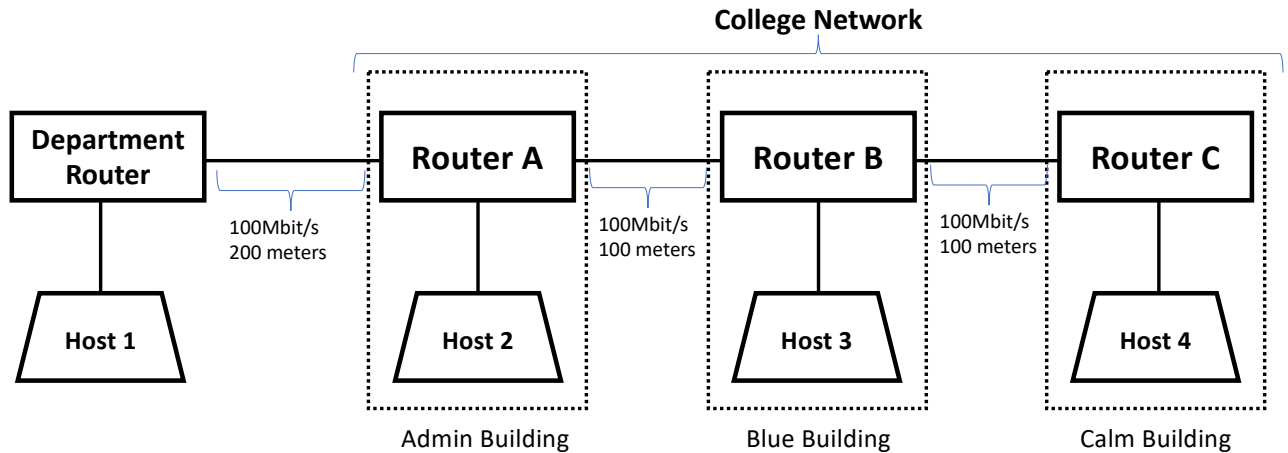


COMPUTER SCIENCE TRIPOS Part IB – 2020 – Paper 5

5 Computer Networking (awm22)



Topic 1 – end-to-end latency; topic 4 IPv4 and IPv6 address resolution, Topic 1 fault finding, cumulative loss

- (a) Using a diagram, illustrate how long a packet of length L bytes will take to travel over an idle network from Host 4 to Host 1? The routers use a store-and-forward architecture. [5 marks]

Answer:

This is an application of the (taught) cause of latency between two end points : Propagation + transmission + queueing delay + per-packet processing

Some values are given within the question but an algebraic statement is required as we do not know the speed of links between host and switch.

Total time = Total Time from Host 4 to Router C +
Total Time between Router C and Router B +
Total Time between Router B and Router A +
Total Time between Router A and Department Router +
Total Time between Department Router and Host 1

each Total Time is a sum of (Propagation + transmission + queueing delay + per-packet processing)

per-packet processing should be stated but may be assumed to be zero

- (b) Computers in Calm building are often not getting allocated IP addresses and the performance is quite poor. The department router serves DHCP for the College network and is operating correctly. Residents in Blue report intermittent performance issues, but no one in Admin reports any problems. Network measurements reveal that the per-router packet loss for each switch under load can be as high as one packet in five thousand, but it is significantly worse for packets smaller than 1000 bytes, where as many as one packet in twenty are lost.

With these insights, explain the cause of the problems experienced. Make clear any simplifying assumptions you have made. [5 marks]

Answer:

The probability of loss cumulates along the path of routers.

The probability of loss accumulates to a significantly higher value for smaller packets $((0.95)^3)$ for packets smaller than 1,000 bytes in length gives (about 15 % loss). [1 mark]

DHCP packets are commonly <1,000 bytes; so subject to a much higher probability of being dropped (lost). [1 mark]

Additionally ACK packets in isolation are small so that would also explain the performance issues too. [1 mark]

finally, ARP is also small packets so a link lossy to small packets adversely affects such operations – however, ARP is local-link only so loss does not cumulate in the same way (ARP doesn't cross routers), but the small packets of ARP will still see a higher loss probability than larger packets for the same link. [1 mark]

an issue here is that DHCP (and ARP) failures are critical; at least ACK loss is recoverable – additionally Probability is not so trivially treated so these numbers/results should be treated with scepticism – it is not “normal”; but the values are insightful – even as bounds. Additionally any treatment assumes many properties that are not valid (independence, etc.) – it is such simplifying assumptions that will gain the last mark.

- (c) Some students in Calm have found using IPv6 will ‘work’ (i.e., connecting to the wider University services is possible, but not to Internet services); although still not performing as well as when they are in the Admin building.

Describe the steps by which IPv6 addresses are allocated without DHCP and consider why this service may be working more reliably than IPv4? [6 marks]

Answer: Describing the method by which addresses are allocated without configuration (aka zero-conf configuration)

Link-Local Address Generation – The device is assigned a link-local address. It comprises of ‘1111111010’ as the first ten bits followed by 54 zeroes and a 64 bit interface identifier.

Link-Local Address Uniqueness Test – In this step, the networked device ensures that the link-local address generated by it is not already used by any other device i.e. the address is tested for its uniqueness.

Link-Local Address Assignment – Once the uniqueness test is cleared, the IP interface is assigned the link local address. The address becomes usable on the local network but not over the Internet.

Router Contact – The networked device makes contact with a local router to determine its next course of action in the auto configuration process.

Router Direction – The node receives specific directions from the router on its next course of action in the auto configuration process.

Global Address Configuration – The host configures itself with its globally unique Internet address. The address comprises of a network prefix provided by the router together with the device identifier. [1 mark each step]

because these processes are router-local (actually interface/segment local) a lack of connectivity to the DHCP services is not relevant. [1 mark]

It is not clear that IPv6 addresses allocated in this way would be routable outside the institution. [1 mark]

- (d) Two approaches to improve the network performance are available: one is to upgrade the performance of the physical links between the buildings to 10Gbit/s. The alternative approach is to significantly change the topology of the network by adding an additional high performance router, but leaving the performance of the physical links unchanged.

— *Solution notes* —

Briefly give the advantages and disadvantages of each approach. [4 marks]

Answer: upgrading the links doesn't upgrade the routers and, as it stands, they are the problem. [1 mark]

Additionally the issue is the cumulative Pr of packet loss along the line of switches. [1 mark]

A sure solution to at least make things fairer through a star topology – in this way all hosts on any switch will share a single local switch, but then share fairly the link to Host 1 [1 mark]

the change in the speed of light for fibre is a point of contrast with the existing latency numbers (from part a) but don't fundamentally change the problem of the cumulative loss probabilities. [1 mark]
